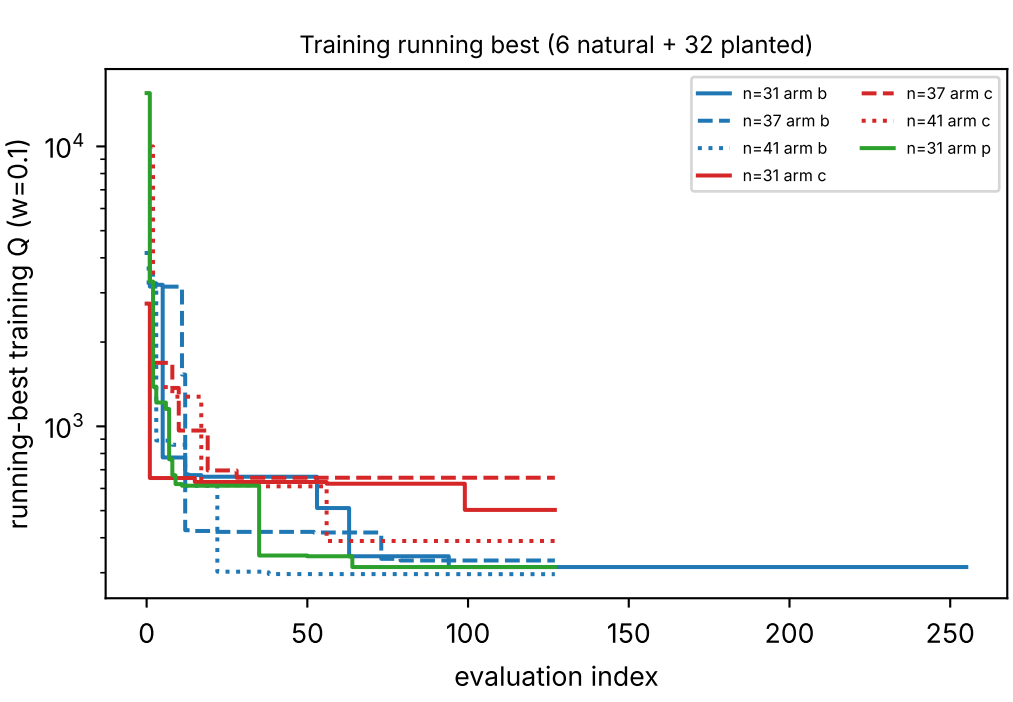
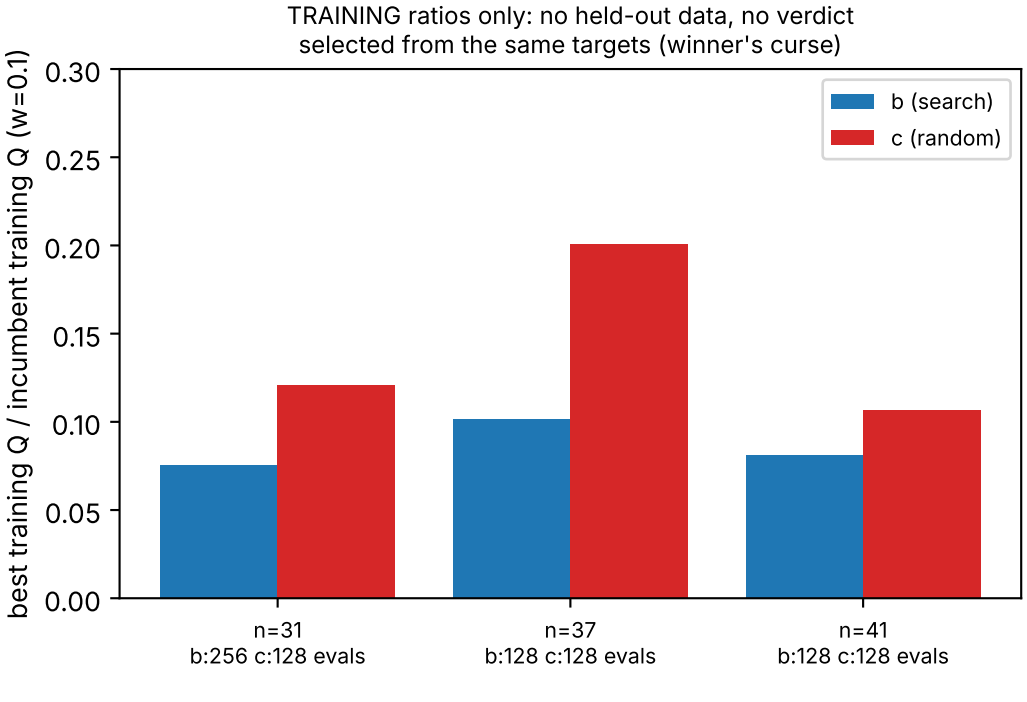
Item 10 kernel program search, stage 1 of 4 complete (CARD-BLINDSPOT-10-02/05/06/07/08/10/12-r2)





Item 10: program search over the decomposition kernel (assessment after repair pass R2)

Verdict: INCONCLUSIVE (level: constant; no verdict can be computed yet)

The frozen decision rule (`PROTOCOL.md`, "Decision rule") is defined on **held-out** data
(128 natural + 128 planted targets per cell, 2000-resample bootstrap intervals for
`Q(B)/Q(A)` and `Q(B)/Q(C)` at w = 0, 0.1, 1). The held-out, finalist, ablation, conservation and
summary cards (14-18, 21, 22) have not run: stages 2-4 are not frozen or executed. Stage 1 of 4 is
complete, and the search archives are only partly complete (see the table). So no per-cell ADVANCES /
MATCHES / REGRESSES / TRADES can be assigned, and the rule's own "otherwise" branch applies.
Nothing here is a speedup claim, and nothing is credited to the search.

Distance to matched rho: **unchanged and not measured by this item.** It is a one-stage diagnostic
(S_4 pair-and-root kernel, toy factor bases, l = 9, n = 31, 37, 41); it computes no S, no rho ratio and
no GAE, so the section 5a pricing band (0, 0.1, 1 GAE) has no cross-method row to rank. The unpriced
native
counters (word XORs, hash probes, trace tests, pair iterations) are priced at 0, 0.1 and 1 counted
operations inside Q for every arm, the same way. Even a full success would be a constant factor on
the relation phase, not the exponent.

First run (superseded)

The first run left no assessment: no README existed. It produced receipts for card 01 (selftest,
completed) and
card 04 (n31 arm b, evaluations 0..191, completed), both kept and not re-run. Cards 02 and 03 were cut
off
without a receipt; a partial `raw/state/n31-p.jsonl` (151 evaluations) has no receipt and is **not used**
(its evaluations 0..127 agree field for field with the R2 card-02 archive, which is consistent with the
evaluations being pure functions of the index; noted, not relied on). Cards 05-18 never ran. No first-run
verdict exists to supersede;
the status was "incomplete, nothing assessable".

What ran (R2 pass, all relays checked against receipts)

card	content	status	archive (sha256 prefix)
01 (first run)	build + selftest	completed	`raw/selftest.txt`
04 (first run)	n31 arm b, evals 0..191	completed	`raw/state/n31-b.jsonl` 23792d0b
02-r2	n31 arm p, evals 0..127	completed	`raw-r2/state-1/n31-p.jsonl` 83fac4f4
05-r2	n31 arm b, evals 192..255 (copy of card 04 archive, bound by sha, then appended)	completed
`raw-r2/state/n31-b.jsonl` 8d4e27e9
06-r2	n37 arm b, evals 0..127	completed	`raw-r2/state-1/n37-b.jsonl` 0c8c0d54
07-r2	n41 arm b, evals 0..127	completed	`raw-r2/state-1/n41-b.jsonl` 59c6c7fc
08-r2	n31 arm c, evals 0..127	completed	`raw-r2/state-1/n31-c.jsonl` e44de45a
10-r2	n37 arm c, evals 0..127	completed	`raw-r2/state-1/n37-c.jsonl` 0154f724
12-r2	n41 arm c, evals 0..127	completed	`raw-r2/state-1/n41-c.jsonl` ebaf9b23

Relay check (`tools/task_card.py check-relay`, reports in `raw/relay-1..7.txt`): cards 02, 06, 07, 08,
10, 12-r2
`relay_verbatim: true`, no findings. **Card 05-r2: check-relay returned `CARD FAIL`, not a relay
finding:**
its `copy-card-04-archive` command names the destination file `raw-r2/state/n31-b.jsonl`, which the card
itself
created, and the linter (run after the fact) now reports it as an unpinned repository file. Checked by
hand
instead: the relayed card sha, receipt sha (e8a6d9a4...) and output sha (8d4e27e9...) equal the on-disk
values,
the bound input `raw/state/n31-b.jsonl` still hashes to 23792d0b..., and the first 192 lines of the new
archive are
byte-identical to it. This is reported as a deviation of the check, not repaired; the data are usable.
Not run yet: cards 03, 19, 20, 09, 11, 13 (stage 2: the second 128 evaluations of p-n31 b-n37 b-n41 c-n31
c-n37 c-n41),

14-18 and 21 (stage 3), 22 (stage 4).

Numbers (training set only, Q at the stated weight; measured, from the archives)

Training = 6 natural + 32 planted targets per base. The incumbent is evaluation 0 of each arm-b archive
(Q at w = 0.1: n31 4162.2, n37 3264.2, n41 3663.4; planted retention phi = 1). "ratio" is the best finite
training Q of
the arm divided by the incumbent's Q on the same targets. These ratios use the targets the programs were
selected on, have no interval, and are exactly the quantity the rule forbids reading as a verdict
(selection among 128-256 candidates on 6 natural targets, so they are optimistically biased).

| cell, arm | evals done / 256 | best program (training) | Q at w=0.1 | ratio at w = 0 / 0.1 / 1 | phi |
dominated, Q=inf |
|:--|:--|:--|:--:|:--:|:--:|
| n31 b | 256 | affine-aligned-s2 lift1 stop0 root0 quartic=b1(1) sqt1 ab=2 bt=1 (idx 94) | 314.3 |
0.0753 / 0.0755 / 0.0767 | 0.94 | 2, 10 |
| n37 b | 128 | affine-aligned-s1 lift1 stop1 root0 q1 sqt1 ab4 bt1 (idx 79) | 331.8 | 0.1014 / 0.1016 /
0.1031 | 0.31 | 3, 5 |
| n41 b | 128 | affine-aligned-s2 lift1 stop1 root0 q1 sqt1 ab4 bt1 (idx 66) | 296.4 | 0.0807 / 0.0809 /
0.0821 | 0.34 | 0, 0 |
| n31 c | 128 | affine-rand-s1 lift1 stop0 root0 q1 sqt0 pm1 ab4 bt1 (idx 99) | 502.9 | 0.1207 / 0.1208 /
0.1219 | 0.44 | 26, 39 |
| n37 c | 128 | sub-low lift1 stop1 root1 q0 ab4 bt1 (idx 28) | 655.4 | 0.1979 / 0.2008 / 0.2172 | 0.50 |
48, 55 |
| n41 c | 128 | affine-aligned-s1 lift1 stop0 root1 q1 sqt1 pm1 ab4 bt2 (idx 56) | 389.5 | 0.1058 /
0.1063 / 0.1092 | 0.53 | 39, 44 |
| n31 p | 128 | affine-aligned-s2 (same program as n31 b idx 94, found at idx 64) | 314.3 | start 15505
-> 314.3 | 0.94 | 0, 0 |

Running-best training Q (w = 0.1, recorded): n31 b 4162 -> 343 (64 evals) -> 314 (128) -> 314 (256); n37
b 3264 -> 418 -> 332;
n41 b 3663 -> 297 -> 296. All three search arms reach their plateau by evaluation 128 or earlier.
Search versus random on equal evaluations (128 each, training, w = 0.1): best-b / best-c = 0.625 (n31, b
at 256),
0.506 (n37), 0.761 (n41). Descriptive only: no interval, no held-out, and arm c is half complete.

Controls (state as of now)

* Instrument correctness (selftest, `raw/selftest.txt`): 400 random programs against brute force at n =
17, l = 6,
  bad = 0; quartic forms equal the resultant; S_4 vanishes on x(P1 +- P2 +- P3) at n = 17, 31, 41; `#E`
and
  `2E = ker delta` at n = 13..41; handicap checks 1.996 (pairs=full, 1.95-2.05), 0.213 (lift, 0.20-0.30),
0.234
  (abort 1/8, 0.22-0.25), 1.542 (generic root > 1.3), 0.9858 (b1 < 1): **all PASS**.
* Positive control for the search (arm p), provisional at 128 of 256 evaluations: best training Q 314.3
is at most the
  incumbent's 4162.2 and at most half of its start 15505. The pass criterion is met so far (ratio 0.0203
to its start),
  but the arm is not finished and the criterion is on the final 256, so it is recorded as "met
provisionally".
* Negative control, arm c: half complete (128 of 256). Its best training Q is also far below the
incumbent's
  (ratios 0.12, 0.20, 0.11), so random search over this program space already finds large training gains;
whether
  the search beats it is exactly the `Q(B)/Q(C)` comparison that needs held-out data.
  `affine-anti` never receives a finite Q wherever it was drawn (arms b n31, b n37 and the c archives):
consistent
  with its prediction.
* Null (incumbent on two held-out halves), yield conservation at (17,6) and (19,7), finalists, ablation:
not run.
  A finalist whose family violates the yield model cannot support a verdict, and that check has not

happened.

Observations that the final held-out stage must test (not findings)

1. The best training programs share `lift = 1` (the incumbent's `lift = 0` scans abscissae without a point):
   the best `sub-low` program in the n31 b archive (lift1, otherwise incumbent-like) already has Q 834 (ratio 0.20).
   The reference arm a1 (incumbent plus `lift = 1, stop = 1`) is a card-14 output and has not run, so how much of the
   gain is "do not scan non-rational abscissae" is **not yet separated**; the rule reports `Q(B)/Q(A1)` for this.
2. The winners are `affine-aligned` bases. In Q this earns `T_eff` credit (n31: 2.90e6 versus 1.11e6; n37: 3.21e6
   versus 1.60e6; n41: 3.10e6 versus 1.35e6, factors 2.6, 2.0, 2.3), which is the yield model, not a cheaper kernel;
   whether the model holds at those bases is the conservation check, which has not run. Part of the training gain
   is therefore contingent on an unrun check.
3. The aligned bases give up planted retention (phi 0.94, 0.31, 0.34 for the three winners): Q divides by phi. The
   held-out phi is what counts.

Deviations and caveats

* Staging, not a protocol change: the R2 pass put all outputs under `raw-r2/` and split searches into 128-evaluation
  pieces across cards. The addendum to `PROTOCOL.md` records this; question, arms, cells, metric and rule unchanged.
* The relay check of card 05-r2 fails only the after-the-fact lint (above).
* The partial first-run `raw/state/n31-p.jsonl` (151 evaluations, no receipt) is not used.
* Host L0 (shared 4-vCPU container, two lanes at once): wall times in the archives are descriptive only; the
  verdict quantity is counted operations. Toy scale: GF(2^n), n = 31..41, l = 9, sparse regime (about 0.02 raw
  S_4 roots per natural target); a replica of the incumbent kernel, not the crate's `SubspaceOracle` (Deviation 1);
  no result here transfers to cryptographic sizes.
* Not a performance result in the sense of AGENTS.md: no held-out measurement exists, so nothing was copied to
  `benchmarks/` and no leaderboard section was regenerated. Re-evaluate after stage 3.

Next steps (not done here)

Commit the stage-1 outputs (this commit), run `make_r2_cards.py 2`, execute stage-2 cards, then `3` and `4`;
then apply the rule to the held-out summary (`raw-r2/summary.json`, card 22) and replace this verdict.

Files

`graph.svg`, `graph.png` (source `make_graph.py`, data `raw-r2/training-stage-summary.json`), `REPORT.pdf`,
`raw/relay-1..7.txt`, `raw-r2/receipt-card-*-r2.json`, `raw-r2/state*/`.